Algebra 1: Second Semester Final Review

Chapters 6, 7, and Rad

1. When multiplying same bases what do you do to the exponents?
2. When dividing same bases what do you do to the exponents?
3. When you have a power raised to another power what do you do to the exponents?
4. Anything raised to the zero power is equal to what?
5. How do you undo a negative power?

Simplify the expression. Write your answer using only positive exponents.

6. $\frac{2^{-3}x^0}{y^{-4}}$

7. $(-4x^2)^3$

8. $5^{-6} \bullet 5^3$

9. 3^0

10. $6xz^{-2} \cdot x^2z$

11. $12c^{-7}d^6$

Solve the equation.

12. $3^{x-2} = 9^{2x+3}$

13. $25^{2x-3} = 25^{x+1}$

14. Define exponential function:

15. Does the following table represent an exponential function?

a)

-2	2
-1	4
0	8
1	16

b)

-2	2
-1	4
0	6
1	8

16. You deposit \$500 in a savings account that earns 7% interest compounded annually. $A = P\left(1 + \frac{r}{n}\right)^{nt}$

A) Write the function that represents the balance after t years.
B) What is the balance after 2 years?

17. You buy a used car for \$6599. Its value decreases by 12% every year.
A) Write the function that represents the value y (in dollars) of the car after t years.
B) What is the value of the car after 20 years?

18. Write the function in $y = ab^x$ form that matches the situation: A population of 520,450 seagulls is increasing 6% yearly.

19. Determine if the function is an exponential growth or decay or neither. $f(t) = 5(.13)^t$

20. Determine if the function is an exponential growth or decay or neither. $f(t) = 3(1.5)^t$

21. Write the radical expression in exponential form: $\sqrt[4]{8}$

22. Simplify each: a) 2^{-1} b) $4^{\frac{1}{2}}$ c) 3^{-3}

Chapter 7

Write the polynomial in standard form. Identify the degree of the polynomial. Then classify the polynomial by the number of terms.

23. $5 + x^2 - 7x$

24. $2x - 4x^5 + 3x^9 - 1 + x^2$

Find the sum or difference.

25. $\left(-5x^2 - 2\right) + \left(3x^2 + 7\right)$

26. $\left(y^2 + 4 - 2y^3\right) - \left(3y^2 - 7y + 4y^3\right)$

27. $(-x + 3) + (-11x^2 - 8 + 12x)$

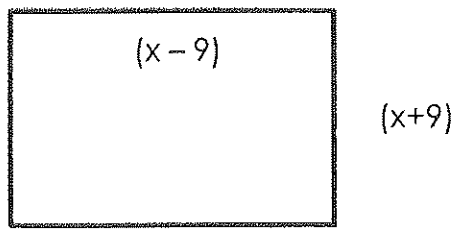
28. $(2b^3 - 5b) - (7b + 3b^2)$

Find the product.

29. $2(x-1)(2x+2)$

30. $(a+b)^2$

31. Find the area of the rectangle.



Simplify:

32. $\sqrt{27}$

33. $\sqrt{24}$

34. $\sqrt{16x^4}$

35. $\frac{5}{\sqrt{5}}$

FACTOR

36. $4s^2 - 1$

37. $x^2 - 3x - 28$

38. $9x^2 - 100$

39. $x^2 - 13x + 40$

40. $a^2 + 10a + 16$

41. $2x^3 - 4x^2 - x + 2$